

Please refer to all relevant safety docs, manufacturers docs, general electrical safety before constructing! Be safe!

Parts List

- **2x 10k Ohm Resistors:** Standard 1/4 Watt through-hole resistors (carbon film or metal film).
- **Dupont Jumper Wires:** To connect the sensor to the breadboard/resistors and the divider to the Cerelog headers.
- **Small Solderless Breadboard:** For the initial build and testing. Once validated, you can solder these inline and wrap them in heat-shrink tubing for a permanent, rugged cable.
- **Digital Multimeter:** Mandatory for verifying the voltage drop before plugging anything into the ADS1299.

Build Directions

Wiring, per sensor:

1. Sensor VCC to the board's 3.3V pin. Sensor GND to a board GND pin. Power the sensor from the Cerelog, not a separate supply — it needs to share the board's ground domain.

2. Put your two 10k resistors in series between the sensor output and ground. Sensor OUT to one end of the first resistor, the other end of the first resistor joins the second, and the far end of the second goes to GND.

3. That junction where the two resistors meet is your center tap. That point — not the sensor OUT pin — is what goes into the board. Run it to the positive input of a spare channel, IN5P say.

4. Tie that channel's negative input, IN5N, to board GND.

The center tap gives you about 1.65V, which the front end handles fine at gain 1. Second sensor goes on IN6 the same way, with its own pair of resistors.

Before you connect anything to the board, put a multimeter on the center tap. You should read around 1.6V with the sensor lit and near 0V with it covered. If you read 3.3V, the divider is wired wrong — fix it before plugging in.

5.Perform the Bench Check:Critical to prevent hardware damage.

Power on the Cerelog board. Turn on your multimeter and set it to measure DC Volts.

- Place the black probe on Cerelog GND.
- Place the red probe on the Center Tap.
- **Test:** Shine a bright light on the ALS-PT19. The multimeter must read roughly **1.6V to 1.65V**.
- **Test:** Cover the sensor with your hand. The multimeter must drop to near **0V**.

If you read 3.3V, your divider is wired incorrectly. Stop and rebuild it.